

Interpreting Powers of i and the Complex Plane

powers of i , plotting on the complex plane, multiplying by i as a quarter turn, and non-real solutions

Directions. Every problem below has two halves: a *picture or pattern* and the *notation* that records it. Do both. On the complex plane, the number $a + bi$ is the point (a, b) — the real part runs along the horizontal (Re) axis and the imaginary part along the vertical (Im) axis. Write every answer in $a + bi$ form.

1. The cycle. Fill in the three empty cells, then answer the question underneath.

i^1	i^2	i^3	i^4	i^5
i	-1			

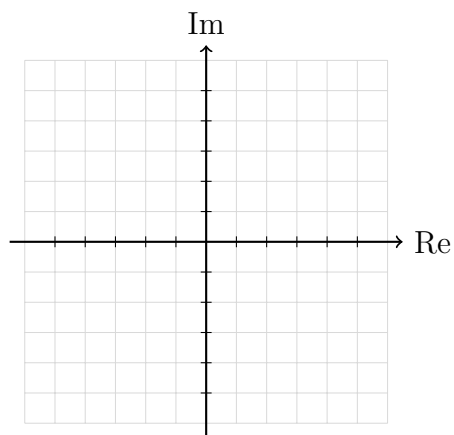
What is i^5 , and why does the table start repeating there?

Answer: _____

2. Using the cycle. Simplify i^{27} . Show the division you used to decide where the exponent 27 lands in the four-step cycle.

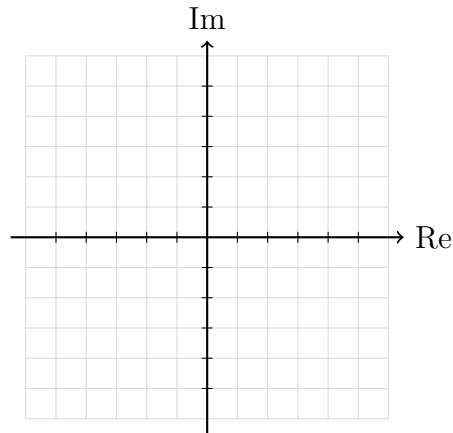
Answer: _____

3. Addition is tip-to-tail. Plot $3 + 2i$ and $-1 + 4i$ as arrows from the origin on the plane below. Slide the second arrow so its tail sits at the tip of the first; the tip now lands on the sum. Write that sum in $a + bi$ form and check it by adding the real and imaginary parts separately.



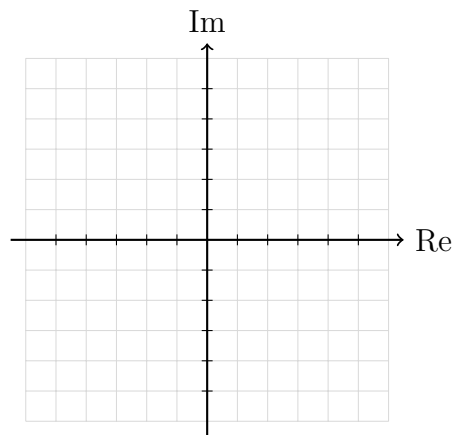
Answer: _____

4. Multiplying by i is a quarter turn. Plot $3 + 4i$. Then compute $i(3 + 4i)$ using $i^2 = -1$, write the result in $a + bi$ form, and plot it too. Describe in one sentence what multiplying by i did to the arrow.



Answer: _____

5. How far from the origin? Plot $3 + 4i$ on the plane below and drop a right triangle from it to the real axis. Use the Pythagorean theorem on the legs to find the distance from the origin to that point.



Answer: _____

6. No real solutions, two complex ones. Solve $x^2 + 9 = 0$. Then say where the two solutions sit on the complex plane and what the picture has to do with the fact that the parabola $y = x^2 + 9$ never crosses the x -axis.

Answer: _____

7. Two powers at once. Simplify $3i^{15} - 2i^{22}$ to $a + bi$ form. Reduce each power with the cycle first, then combine.

Answer: _____

8. Four quarter turns. Start at $1 + 2i$ and multiply by i four times in a row — that is, compute $i^4(1 + 2i)$. Write where you land, and explain why the picture guarantees that answer before you do any arithmetic.

Answer: _____

9. All the zeros. Find all four zeros of $x^4 - 16$. (Hint: factor as a difference of squares twice.) Say which zeros are real and which are not, and describe the pattern the four points make around the origin.

Answer: _____

10. Conjugates. Solve $x^2 - 4x + 13 = 0$ by completing the square or with the quadratic formula. Plot both solutions mentally: they are mirror images of each other across the real axis. Explain why that mirror symmetry has to happen for any quadratic with real coefficients.

Answer: _____